GR CARBONS

Activated carbon plant

User manual

This manual is for people who receive raw material, run the kilns and destoners, screen and finish the carbon, and dispatch bags. It follows the screens in Nyra MES.

Issued 16 October 2026. Your menu shows only the areas your role can open.

1. How a bag moves through the plant

Material is received at the gate, prepared, and released. A work order then says what grade, CTC, and kilograms the kiln should make. After the kiln, destoning checks for stones. Clean bags go to screening and the later steps. A bag stops when Quality sets the next destination to In Stock. Dispatch ships from In Stock.

Stage	Where you work	What you leave behind
Receive	Raw Material	An inward number, ORD-…
Prepare	Raw Material → Inbound prep	RMS bags, then outward bags
Release	Raw Material → Release to production	Bags ready for the kiln
Activate	Execution → Activation	Kiln bags and destoner bags
Finish	Execution → Post activation	Screening and later bags
Hold	In Stock	Bags waiting to be shipped
Ship	Dispatch	A dispatch against a work order

There is no Finished Goods step. When a bag is finished, Quality sends it to In Stock.

2. Sign in and the names you see

Sign in with the user id you were given, for example rms@grcarbons. The Dashboard opens first. Use Show if you need to check the password as you type. Reset password emails a code to the address on the user, then you set a new password.

The top right of the screen shows your role: Administrator, Plant Manager, Raw-material section, Plant Entry, Activation, Post Activation, or Quality Control. If a nickname has been set, the nickname is shown and the role sits under it. Lists that say who saved a reading use the nickname. The record still stores the user id.

- A nickname is unique in the plant. Two people cannot use the same nickname, even if the capitals differ.
- An administrator creates a user under Admin → Users, enters a temporary password, and that password is emailed. The person signs in once with it and then chooses their own password.
- Your menu shows only the areas your role can open.

Who does the work

Work	Who saves it
Gate receipt	Plant Manager, Production Planner, Administrator, or Plant Entry
Bag generation	Raw-material section

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Work	Who saves it
Lab	Quality Control
Debit note	Plant Manager, Production Planner, or Administrator
RMS outward	Raw-material section
Release to production	Raw-material section, Plant Manager, or Production Planner
Kiln feed, kiln output, destoning	Activation. A shop-floor operator can run the machines.
Kiln output quality, destoning quality, post-activation quality	Quality Control. A Plant Manager or Production Planner can also open these.
Screening through blending	Post Activation. A shop-floor operator can run the machines.
Destoner CTC allocation and Stock Reallocation	Plant Manager, Production Planner, or Administrator
Dispatch	Plant Manager or Production Planner

The left menu

Menu	Use it to
Dashboard	See the plant, machine use, and trace a bag or work order
Raw Material	Gate receipt, inbound prep, release, and raw-material reports
Execution → Activation	Kiln feed, kiln output, destoning, and kiln readings
Execution → Post activation	Screening through blending, then quality
Dispatch	Ship In Stock bags for a work order
Reports	Read what was done. Reports do not change the bags
Admin	Users, suppliers, Destoner CTC allocation, kiln settings, work orders, Stock Reallocation
Masters	Plant rules, mesh, machines, and inbound settings. Nyra only.

User ids, passwords, and the screens each login can open are on the separate plant logins sheet.

3. Numbers on bags and orders

Each number has one job. Use the number printed on the bag or shown on the screen. Do not invent a second number for the same bag.

Number	Meaning
ORD-…	One gate receipt. This is the inward until it is released.
WO-…	A work order: grade, CTC, and kilograms to make.
RI_BAG_…	A raw-material bag made after the gate weight is saved.
RO_BAG_…	An outward bag after rotary preparation. This is what the kiln can load.
KA_… or KB_…	A kiln output bag. KA is Kiln A. KB is Kiln B. The middle is the date.
DS_…	A destoner bag.
SC_, DD_, CR_, DM_, BL_	Screening, de-dusting, crushing, de-magnetisation, blending.
DN-…	A debit note raised after lab, when one is needed.

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The date in a bag number is the India date, written day-month-year. Example: 121026 is 12 October 2026. A bag keeps the date it was created. It does not change later in the day.

4. Raw material

Gate receipt

Plant Entry and the supervisor open Raw Material → Gate receipt. The Raw-material section login does not see this tab. Record one inward for the truck or delivery you are receiving. Saving does not create bags.

1. Choose the supplier.
2. Enter the DC number, the supplier weight in kilograms, and our weight in kilograms.
3. Our weight is the quantity the plant received. It must be greater than zero and not more than the supplier weight.
4. Save the inward. The screen gives it an ORD number.

Inbound prep

RMS inward bags

The Raw-material section login opens Inbound prep → Bag generation. Type each bag weight and click Generate bag. Keep adding bags until the bag total matches our weight. Bags are numbered RI_BAG_…. The system does not split the weight for you.

If the bags are still short of our weight, the screen asks you to confirm before you go on to Lab.

Lab

Quality Control opens Lab and enters moisture, dust, and volatility, each as a percent from 0 to 100, then approves or denies. The values stay on the screen after save. The Raw-material section and the supervisor do not save Lab. Weights stay in kilograms. Deny stops this inward.

A debit note is optional and is raised by the supervisor after lab, when our weight is less than the supplier weight or when moisture is above the plant limit (20 percent unless Masters has changed it). Enter the debit value. The note is numbered DN-…. It does not stop RMS outward. Quality Control and the Raw-material section do not raise it.

RMS outward

The Raw-material section login creates the rotary output bags after Lab. Choose Rotary A or Rotary B and add at least one line. A grade, Unburnt, or Stones each counts. The total cannot be more than the inward weight. When the outward is finished, click RMS outward complete. The inward is then ready for release. You do not need a separate raw-material quality step.

Release to production

The Raw-material section, the Plant Manager, or the Production Planner opens Release to production and releases the inward. Release waits until RMS outward is complete. Released outward bags can be loaded on the kiln. If the inward is already released, the release button stays off. After release, go to Execution.

5. Work orders

A work order is the production demand. Open Admin → Work orders. Enter the grade, the CTC, and the kilograms. The screen numbers it WO-…. CTC is entered once here. Later screens show that CTC. They do not ask you to type it again, except kiln output quality, which measures the actual CTC of the bag that came out of the kiln.

Create the work order before kiln feed. The kiln operator picks this number. It is not a customer order. Customer orders are not recorded in this plant.

Stock Reallocation, under Admin, moves an In Stock bag onto another work order of the same grade. The bag keeps its number and its earlier work order in the history. A different grade, the same work order, or a bag that is not In Stock is refused.

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6. Activation

Open Execution. The header shows the flow Kiln Feed → Kiln Output → Destoning, and a line for kiln parameters. Choose the step from that flow.

Kiln feed

1 Pick the work order number, then the kiln.
2 A kiln may hold more than one open work order when the grade and CTC are the same. A different grade or CTC waits until the feed already on that kiln has been covered by output.
3 Load released outward bags. A bag already loaded is not offered again. Unburnt and Stones stay off the kiln. The outward bag does not have to match the work-order grade.
4 A work order leaves the open list once kiln output covers its quantity.
5 Feed can continue while output is taken. You do not wait for the kiln to empty before the next load.

Kiln output

Enter the weight only. You do not pick the work order on this screen. The bag is tagged to an open work order on that kiln that still has feed left, using the earliest feed first. Output cannot be more than what is still in the kiln (everything loaded so far, minus everything already taken out). If nothing is loaded, output is blocked. Kiln A bags start with KA_. Kiln B bags start with KB_. The current kiln settings are shown on this screen and are stored on the new bag.

Quality Control records kiln output quality, including the actual CTC. Activation does not open that form. If the measured CTC matches the work order, that weight counts toward the work order. If it does not match, the bag stays for destoning by the measured CTC and does not fulfill the work order.

Kiln parameters

On Activation, Kiln Parameters is where the engineer records T1, T2, T3, Main Damper (MD), Boiler Damper (BD), and Boiler Pressure every 30 minutes. A missed interval is locked until a plant manager approves it, and then the values can be entered. The standing settings (residence time, temperatures, steam, oxygen, pressure) are kept under Admin → Kiln parameters. Those settings stay until someone changes them. They are not entered again every day. Each save is kept in the change log, with who saved it and when. A new kiln output bag keeps the settings that were current when it was created. Traceability shows those settings on that bag. Scrap is not a kiln parameter. Residence time also sets the designed monthly kilograms on the dashboard: a shorter residence raises the designed output, a longer residence lowers it. Other machines keep the capacity entered in Masters.

Destoning

Before the shift loads a destoner, a supervisor or administrator sets today’s CTC under Admin → Destoner CTC allocation. There are four destoners: Destoner 1, Destoner 2, Destoner 3, and Destoner 4. Load bag accepts a kiln bag whose CTC matches that destoner’s CTC for today.

Quality Control records destoning quality. It is a stones check. Stones Yes sends the bag back to destoning. Stones No sends it to Screening. Destoning does not send a bag to In Stock.

7. Post activation

Open Execution. The header says Screening through blending, then In Stock. Each process has a loader. Load a bag, then tag the output weight. Output weight cannot be more than the kilograms still on the loader (in, minus out). The output list on the shop floor shows bags from the current plant day. Earlier days stay on the dashboard and in reports. The loader still shows the bags that are on the machine.

Screening

1 Choose the mesh first. Meshes are A to E, as set in Masters.
2 Load bag stays off until a mesh is chosen.
3 That mesh stays while bags are on the loader. Unload every bag before you choose a different mesh.

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For plant use. The corner shows your role. The sign-in name stays the user id.

4 Tag only a grade that belongs to that mesh. Enter the weight and click Tag.

Later processes

- De-dusting and de-magnetisation use the grade Exklin.
- Crushing uses the crushing grades shown on the screen.
- Blending follows the same load, then tag, pattern.
- The work order number and its CTC are shown. Do not type the CTC again.

Quality and the next destination

Quality Control opens Quality. Activation and Post Activation do not save it. The list shows output bags from the current plant day. Set Next Destination and save. The bag then waits on that process's Load bag list.

- In Stock — the bag is finished. It leaves the process line and can be dispatched.
- Screening, De-dusting, Crushing, De-magnetize, or Blending — load the bag on that process next.

A saved quality result cannot be edited. If the destination is wrong, do not try to overwrite it. Ask a supervisor before any correction outside this screen.

8. In Stock and dispatch

In Stock is not a machine. It is the list of bags whose quality destination is In Stock and that still have kilograms left. Dispatch draws from this list.

1 Open Dispatch.
2 Pick the work order number.
3 The list shows In Stock bags on that work order.
4 Tick the In Stock bags to ship, or use Select all. The screen shows dispatched kg, pending kg, and the kg you have selected.
5 Dispatch uses the selected bags. It does not take kilograms from a bag you left unticked, and it cannot be more than the pending work-order quantity.
6 Shipped kilograms come off the bag. A bag that is fully shipped leaves the available list.

9. Looking back

Dashboard

The plant dashboard opens with three dials: plant yield, overall process efficiency, and supplier efficiency. The figures use all records to date. Machine output is the current month. The view refreshes every 10 minutes. Plant yield is finished kilograms divided by raw material charged. Overall process efficiency subtracts finished loss from 100. A stream still in process stays off that number. Supplier efficiency is 0.5 × recovery + 0.3 × lab pass + 0.2 × weigh score. Choosing a dial or a bar opens that area. Material in plant is charcoal, awaiting kiln, in process, and in stock. Process Tracking holds mesh compliance, destoner CTC, deviation, and the efficiency detail. Kiln designed output follows the residence time. The other machines use the monthly capacity from Masters. The arithmetic for each dial is on the separate calculation formulas sheet. These charts do not change a bag.

Traceability

On the Dashboard, open Traceability. Look up a work order, a bag, an inward, or a dispatch from this month's list. The screen walks back to the gate receipt and the supplier. If more than one supplier fed the work order, the screen says so and shows each supplier's feed kilograms. A kiln output bag also shows the kiln settings stored on it. The customer is not shown.

Reports

Open Reports on Raw Material, on Activation, or on Post Activation. Pick From and To. The calendar closes when the date is chosen. The table lists the bags and readings in that range. Type in Search to narrow the rows, for example

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a bag number or a work order. You can export to Excel. Reports are read-only.

- Supplier name and debit-note rows are shown to the Plant Manager, the Production Planner, the Administrator, and the super administrator. Other logins see the weights and bag numbers without the supplier.
- User performance lists who saved the readings. It is shown to the Plant Manager, the Production Planner, the Administrator, and a read-only viewer.
- Plant Production reports, for the Plant Manager and a viewer, are the summary charts: work in process, raw-material loss, kiln yield, and grade stock.
- Plant Entry does not have a Reports tab. The Raw-material section uses Raw Material → Reports.

10. Setup that the shop floor depends on

Who	Where	When
Supervisor	Admin → Destoner CTC allocation	Each day, before destoner load. One CTC per destoner.
Supervisor	Admin → Kiln parameters	When the standing kiln settings change. The change log keeps who saved each setting. Scrap is not a kiln parameter.
Supervisor	Admin → Stock Reallocation	When an In Stock bag must move to another work order of the same grade.
Planner	Admin → Work orders	Before the kiln is fed for that demand.
Admin	Admin → Users	When a person joins, or when a nickname must change.
Admin	Admin → Suppliers	When a new supplier delivers to the gate.
Admin	Masters → Mesh	When the grades on mesh A–E change.
Nyra	Masters → Process rules	Moisture, CTC decimals, bag rules, and minutes from load to output for a 500 kg bag.
Admin	Masters → Inbound settings	Rotary names and the outward grades, including Unburnt and Stones.

11. When a button stays off

- Release stays off until RMS outward is complete, and again after the inward is already released. An administrator login does not save release.
- Kiln output stays off when the kiln has no feed left, or when the weight is larger than what is still in the kiln.
- Destoner Load bag stays off when today’s CTC is not set, or when the bag CTC does not match that destoner.
- Screening Load bag stays off until you choose a mesh.
- A quality row stays locked after it is saved.
- If your menu is missing a screen, your role does not include it. Ask an admin. Do not share another person’s sign-in.

12. A short day on the floor

1. Gate: Plant Entry or a supervisor saves the inward with DC number, supplier weight, and our weight. Saving does not create bags.
2. Raw-material section: generate each bag by kilograms. Quality Control: lab. Raw-material section: at least one outward line, then RMS outward complete.
3. Raw-material section, Plant Manager, or Production Planner: release to production.
4. Confirm today’s Destoner CTC allocation, and that the work order exists.
5. Activation: follow Kiln Feed → Kiln Output → Destoning. Load outward bags, take kiln output by weight (the current kiln settings are stored on that bag), then destone.
6. Quality Control: kiln output quality and destoning quality. Stones No goes to screening.
7. Post Activation: screening, mesh first, then load, then tag the grades on that mesh.

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8 Quality Control: send the bag to the next process, or to In Stock.
9 Plant Manager or Production Planner: dispatch the selected In Stock bags for that work order.

If a screen and this manual disagree, follow the screen and tell the supervisor. The screen is the live rule for that bag.

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